

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Schema Analysis and Viva Voce

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO2: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)

Assessment Tool Weightage: 35%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 2

Total Marks: 20

Code of Conduct

I P.Gunasri (192324103) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Gunasri

Database Design and Connectivity Rubric

Requirement Analysis (4 Marks)	ER Diagram Design (4 Marks)	Table Design & Normalization (4 Marks)	Database Connectivity using Python (4 Marks)	CRUD Operations (4 Marks)	Total (20 Marks)

Total Marks Awarded:

Questions:

Question 5: University Course Registration Database (10 Marks)

A university maintains the following database schema:

- Student (Student_ID, Student_Name, Department)
- Course (Course_ID, Course_Name, Credits)
- Enrollment (Enrollment_ID, Student_ID, Course_ID, Semester)

Analyze the schema and write a Python program to retrieve the student name, course name, and semester for all students enrolled in courses carrying more than 3 credits.

Set B

Question 6: Banking Database (10 Marks)

A bank maintains the following database schema:

- Customer (Customer_ID, Customer_Name, City)
- Account (Account_No, Customer_ID, Account_Type, Balance)
- Transaction (Transaction_ID, Account_No, Transaction_Date, Amount)

Analyze the schema and write a Python program to retrieve the customer name, account number, and balance for customers whose account balance exceeds ₹1,00,000.

Signature of the Course Faculty

Question 5: University Course Registration Database

```
import sqlite3

# Connect to SQLite database
conn = sqlite3.connect("university.db")
cursor = conn.cursor()

# Create Student table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Student(
    Student_ID INTEGER PRIMARY KEY,
    Student_Name TEXT,
    Department TEXT
)
""")

# Create Course table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Course(
    Course_ID INTEGER PRIMARY KEY,
    Course_Name TEXT,
    Credits INTEGER
)
""")

# Create Enrollment table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Enrollment(
    Enrollment_ID INTEGER PRIMARY KEY,
    Student_ID INTEGER,
    Course_ID INTEGER,
    Semester TEXT
)
""")

# Insert sample data
cursor.execute("INSERT INTO Student VALUES (1,'Rahul','CSE')")
cursor.execute("INSERT INTO Student VALUES (2,'Priya','ECE')")
cursor.execute("INSERT INTO Student VALUES (3,'Amit','IT')")

cursor.execute("INSERT INTO Course VALUES (101,'Python Programming',4)")
cursor.execute("INSERT INTO Course VALUES (102,'Mathematics',3)")
cursor.execute("INSERT INTO Course VALUES (103,'Database Systems',5)")

cursor.execute("INSERT INTO Enrollment VALUES (1,1,101,'Semester 3')")
cursor.execute("INSERT INTO Enrollment VALUES (2,2,102,'Semester 2')")
```

```
cursor.execute("INSERT INTO Enrollment VALUES (3,3,103,'Semester 4')")
```

```
conn.commit()
```

```
# Query
```

```
cursor.execute("""
SELECT Student.Student_Name,
       Course.Course_Name,
       Enrollment.Semester
FROM Student
JOIN Enrollment
ON Student.Student_ID = Enrollment.Student_ID
JOIN Course
ON Enrollment.Course_ID = Course.Course_ID
WHERE Course.Credits > 3;
""")
```

```
print("Students enrolled in courses with more than 3 credits:\n")
```

```
for row in cursor.fetchall():
    print("Student Name :", row[0])
    print("Course Name  :", row[1])
    print("Semester    :", row[2])
    print()
```

```
conn.close()
```

```
Students enrolled in courses with more than 3 credits:
```

```
Student Name : Rahul
Course Name  : Python Programming
Semester    : Semester 3
```

```
Student Name : Amit
Course Name  : Database Systems
Semester    : Semester 4
```

Question 6: Banking Database

```
import sqlite3

# Connect to SQLite database
conn = sqlite3.connect("bank.db")
cursor = conn.cursor()

# Create Customer table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Customer(
    Customer_ID INTEGER PRIMARY KEY,
    Customer_Name TEXT,
    City TEXT
)
""")

# Create Account table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Account(
    Account_No INTEGER PRIMARY KEY,
    Customer_ID INTEGER,
    Account_Type TEXT,
    Balance REAL
)
""")

# Create Transaction table
cursor.execute("""
CREATE TABLE IF NOT EXISTS Transactions(
    Transaction_ID INTEGER PRIMARY KEY,
    Account_No INTEGER,
    Transaction_Date TEXT,
    Amount REAL
)
""")

# Insert sample data
cursor.execute("INSERT INTO Customer VALUES (1,'Arun','Chennai')")
cursor.execute("INSERT INTO Customer VALUES (2,'Sneha','Bangalore')")
cursor.execute("INSERT INTO Customer VALUES (3,'Kiran','Hyderabad')")

cursor.execute("INSERT INTO Account VALUES (5001,1,'Savings',150000)")
cursor.execute("INSERT INTO Account VALUES (5002,2,'Current',80000)")
cursor.execute("INSERT INTO Account VALUES (5003,3,'Savings',250000)")
```

```
cursor.execute("INSERT INTO Transactions VALUES (1,5001,'2026-08-01',10000)")
cursor.execute("INSERT INTO Transactions VALUES (2,5002,'2026-08-02',5000)")
cursor.execute("INSERT INTO Transactions VALUES (3,5003,'2026-08-03',20000)")
```

```
conn.commit()
```

```
# Query
```

```
cursor.execute("""
SELECT Customer.Customer_Name,
       Account.Account_No,
       Account.Balance
FROM Customer
JOIN Account
ON Customer.Customer_ID = Account.Customer_ID
WHERE Account.Balance > 100000;
""")
```

```
print("Customers with balance greater than ₹1,00,000:\n")
```

```
for row in cursor.fetchall():
    print("Customer Name :", row[0])
    print("Account Number:", row[1])
    print("Balance      : ₹", row[2])
    print()
```

```
conn.close()
```

```
Customers with balance greater than ₹1,00,000:
```

```
Customer Name : Arun
```

```
Account Number: 5001
```

```
Balance      : ₹ 150000.0
```

```
Customer Name : Kiran
```

```
Account Number: 5003
```

```
Balance      : ₹ 250000.0
```